

Table of contents

Potential is “intrinsic energy of inertia”

elec. Potential structure:

$$U_e := \Phi \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\epsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\epsilon_0} It$$

magn. Potential structure:

$$U_m := A \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\mu_0} I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\mu_0} Qv = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2} \frac{1}{\epsilon_0} vQ =$$

$$\frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2} \frac{1}{\epsilon_0} Itv = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2} \frac{1}{\epsilon_0} Ir = \frac{1}{c^2} vU_e$$

Electric endogen energy $E_{pot} = mc^2$:

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (1)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{m}{n} \frac{c^2}{e} = \frac{m}{q} c^2 = \kappa c^2 \quad (2)$$

Electric induced energy $E_{kin} = \frac{1}{2}mv^2$:

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \epsilon_0 \mathcal{E}^2 \quad (3)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (4)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} CU^2 \quad (5)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} LI^2 \quad (6)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (7)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \epsilon_0 \quad (8)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (9)$$

Energy conservation for (large extended) conductive medium if no forces act on it:

$$F = \int \left(\frac{\partial}{\partial t} (we + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \quad (10)$$

$$\stackrel{lowfrequency}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) + (RI^2 - VI) = 0$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (11)$$

V = Potentialdifference I = Current R = Resistance j = Current density

L = Inductance C = Capacitance Q = charge (ion) $\Psi = LI$ = mag. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector = radiation intensity

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$ = dissipation engergy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor}$ $r = \frac{d}{\sqrt{2}i}\rho = \frac{d}{1+i}i\rho = \text{radius (extension) of conductor}$

$$\rho = kr = \text{"radius of curvature } k\text{"} \quad \omega = 2\pi\nu = k\lambda\nu = \frac{1}{\sqrt{LC}} = \text{frequency}$$

$$\mu = \text{permeability of conductor} \quad \sigma = \text{conductivity of conductor}$$

Low Frequency:

$$\begin{aligned} \omega \ll \frac{1}{\mu\sigma} \frac{2}{r^2} &\iff r \ll d \iff \rho \ll 1 \\ L \approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0 \end{aligned} \quad (12)$$

High Frequency:

$$\begin{aligned} \omega \gg \frac{1}{\mu\sigma} \frac{2}{r^2} &\iff r \gg d \iff \rho \gg 1 \\ L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0 \end{aligned} \quad (13)$$

Energy:

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$\begin{aligned} h = \tan(\alpha) &= \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} &= \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k\lambda \end{aligned} \quad (14)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (15)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (16)$$

$$V_0 = \text{Opposing Potential } (I = 0)$$

¹Wirkungsquantum

$$\alpha = \angle(E_{kin}, \nu) \quad e = \text{electron charge} \quad Q = \text{ion charge}$$

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

$\Rightarrow$ Ionisation Energy of chemical element